Problem 2

Until Oxygen Runs Out

50 mins

Contents

1 1. Scenario **1**

2 2. Problems **1**

2.1 2.1 Light attenuation 2

2.2 2.2 Ambient light from double-slit interference 2

2.3 2.3 Depth determination and ascent model 2

3 3. Appendix: Additional Information **2**

3.1 3.1 Light attenuation 2

3.2 3.2 Double-slit interference 2

3.3 3.3 Fluid behaviour 3

1 1. Scenario

You are a researcher investigating light propagation in a highly scattering and absorbing medium. During a field experiment, a malfunction isolates you inside a submerged test chamber. Instrument panels display the following:

1. A timer shows **50 minutes** of reserve power remaining.
2. A data feed provides measurements of light intensity versus depth from a previous calibration run.
3. A secondary optical setup produces a double-slit interference pattern from ambient illumination outside the chamber. To overcome the highly diffuse, scattered nature of the ambient light, it is gathered through a narrow collimating tube and lens system to ensure spatial coherence before reaching the slits. The pattern's fringe spacing has been recorded, along with the maximum intensity.

Your goal is to determine your current depth using only the supplied data. *(Note: While data files are provided, this problem is designed so that all parts can be fully solved either computationally or analytically by hand.)*

2 2. Problems

2.1 2.1 Light attenuation

1. Plot $\ln I$ versus depth h.
2. Find the attenuation coefficient A and the surface intensity I_0. Write an expression for depth as a function of intensity.

2.2 2.2 Ambient light from double-slit interference

From an interferometer experiment, you have the fringe spacing Δy, slit separation d, screen distance L, wavelength λ, and the maximum intensity I_{max} of the interference pattern.

1. The maximum intensity of a double-slit pattern (with both slits equally illuminated) is $I_{\text{max}} = 4I_{\text{sat}}$, where I_{sat} is the intensity from one slit. Use this to find the ambient intensity I_{amb} at your depth.

2.3 2.3 Depth determination and ascent model

1. Combine the results of tasks 1 and 2 to compute your current depth h_{current}.
2. The fluid in this environment does not obey ordinary buoyancy; motion is difficult at depth but becomes easier as you rise. Using `fluid.json`, determine the functional relationship between the resistive force (or the "difficulty" of ascent) and height above the bottom.
3. Assume you can exert a constant propulsive force. Derive an expression for the time required to ascend from h_{current} to the surface. Evaluate this time numerically and compare it with the **50-minute** reserve.

3 3. Appendix: Additional Information

3.1 3.1 Light attenuation

The Beer–Lambert law $I = I_0 e^{-\alpha x}$ describes intensity decay in many absorbing media. The data in the file may or may not follow this form; you are to determine the actual relationship.

3.2 3.2 Double-slit interference

The small-angle approximation is valid because $L \gg d$. The intensity pattern for two slits illuminated by monochromatic light of wavelength λ is

$$I(\theta) = 4I_0 \cos^2\left(\frac{\pi d \sin\theta}{\lambda}\right),$$

where I_0 is the intensity from one slit. The maxima occur when $\sin\theta = m\lambda/d$, leading to fringe spacing $\Delta y = \lambda L/d$ for small angles.

3.3 3.3 Fluid behaviour

The fluid in this world exhibits a height-dependent effective viscosity or drag coefficient. The data file `fluid.json` provides measurements of the relevant parameter at various heights. Use these to infer the law governing your ascent.